STICS crop model: general presentation

ALIMENTATION
AGRICULTURE
ENVIRONNEMENT

INRA

1. Introduction : history, aim, structure and types of application.

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on benhalf of the EPS (STICS project team)

ALIMENTATION
AGRICULTURE
ENVIRONNEMENT

Crop model structure

(Bonhomme *et al.*, 1996)

duration of
development phases

phenology

- Photosynthesis
- Growth of different organs
- Yield accumulation

Dry matter production
and distribution

constraints
-water
-mineral
- Pest and desesases

Constraintes due to
environment

Origin and strategy of STICS

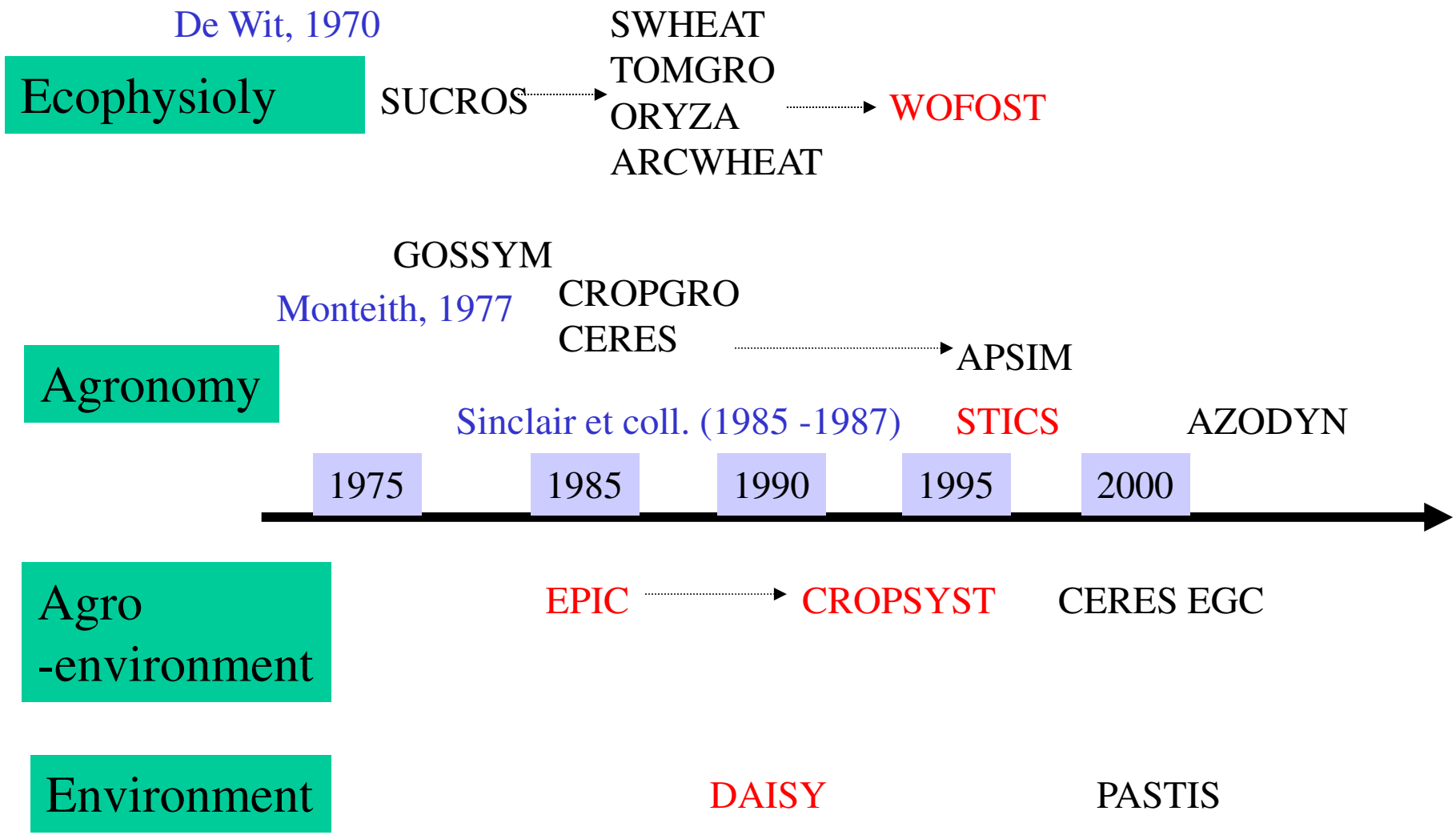
originated from the ECOSPACE project (1996), called Heterogeneities of cultivated areas and agricultural nitrate pollution :

"A **single** model (STICS) .... for different crops. Must integrate knowledge of specialists and generalists in order to remain general, **robust, simple, operational, and flexible** "(GA in October 1996).

=> deterministic functional soil-crop model to describe the thermal balance, radiative, water, C and N across the crop cycle and pedon (grain modeled).

=> integrative modelling of the cropping system functionning.

History and dynamic of STICS



Collective dynamics visible each STICS seminars : Scientific presentations and round tables, exchanges of experience and discussions and decisions on the evolution of the model

	1996	1998	2000	2003	2005	2007	2010	2012
	meeting 1: Laon	meeting 2 Paris				meeting 7 Reims	meeting 8 Toulouse	meeting 9 Orléans
Process ***plante***	Growth & development,							
Process	Water and N balances	Trophic competition, Radiative transfer, Resistive balance						
soil	transfer water, mineralization N, N leaching							
Cultures / practrices	rotations	annuals intercrops (1)						
	horticulture							

Rapid STICS evolution due to several project needs

meeting 1: Laon

meeting 2
Paris

meeting 7
Reims

meeting 8
Toulouse

meeting 9
Orléans

1996 | 1998 | 2000 | 2003 | 2005 | 2007 | 2010 | 2012

Process

plante

Growth & development,

Over-wintering conditions, Senescence

Symbiotic N fixation
Instantaneous NNI
Trophic link between above/belowground DM

N effect on root distribution,
Buds dormancy
Mushrooms
Pathogens

Trophic competition,
Radiative transfer,
Resistive balance

Water and N balances

soil

transfer water,
mineralization N,
N leaching

excess water, mulch,
ammonia volatilization,
nitrification

compaction, fragmentation

Mulch decomposition ,
N2O emission

Cultures / practices

rotations

annuals
intercrops (1)

horticulture

Intercrops (2)

perennials

Mediterranean grassland
Decision planting / harvest

Main traits of the model

is a crop simulation model with agronomical and environmental vocation

Simulated object = one agricultural condition
(set of 1 crop cycle + 1 environmental condition
+ 1 set of agricultural practices) = simulation unit « usm »

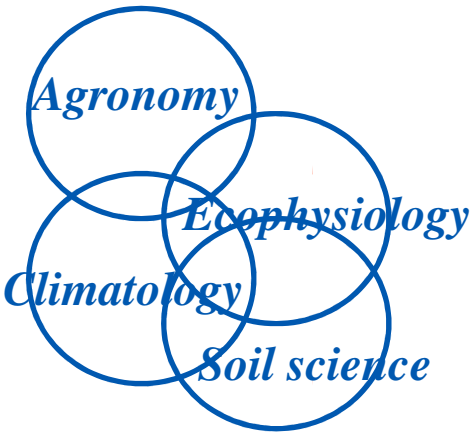
Simulated processes = crop growth and development, energy, water and nitrogen balance

Ability for successive usm linkage or GIS coupling

scientific motivation = operational +
gathering of knowledge + hypothesis testing

Structure

Disciplines

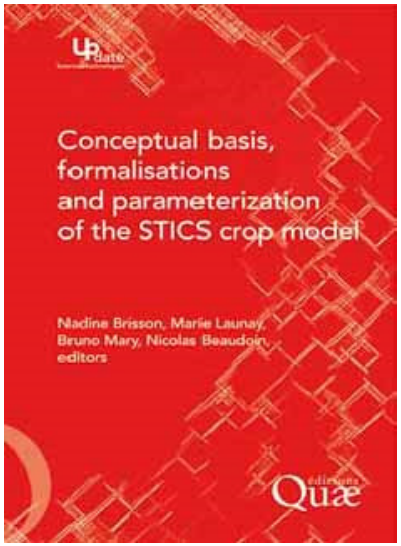


Themes, purposes

- Global change, environment
- New cropping systems

Autors

> 50 scientists



Productions

> 150 citations (1998)

> 150 international papers (2012)

> 30 PhD of which 10 in 2008-2010

> 250 downloads (> 2008)

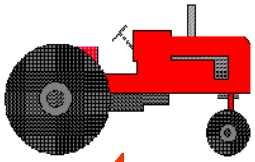
Structure

Crop generality

Plant

wheat, maize
soybean, sorghum
grassland, tomato
sugarbeet, pea
sunflower
vineyard
rapeseeed
banana, carrot
lettuce, sugarcane

Management



crop

Formalisation

sowing

harvest

intercrop

Agricultural variables

Environmental variables

Structure

Crop generality

Plant

wheat, maize
soybean, sorghum
grassland, tomato
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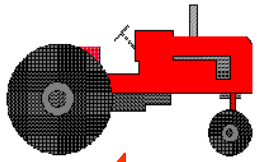
Management

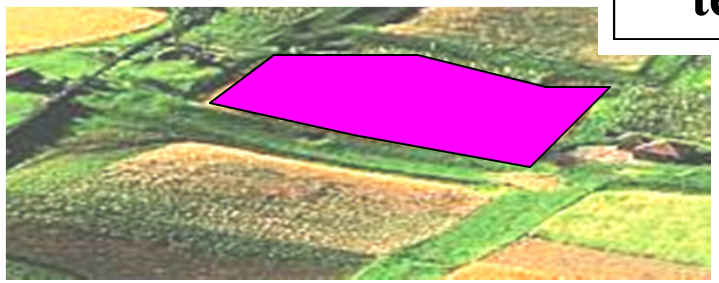


Formalisation

crop

sowing

intercrop

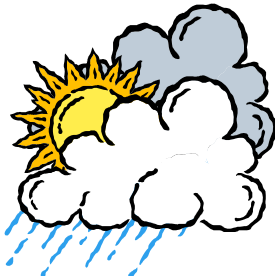


Conceptual modularity

- Phenology
- Shoot growth
- Yield formation
- Crop management
- Microclimate
- Root growth
- Water balance
- Nitrogen balance
- Water, nitrogen and temperature transfers

Conceptual modularity

Structure



Climate

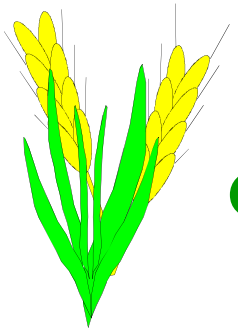
Development

Leaf area index

Radiation absorption

Biomass growth

Harvested organs



Crop

Roots

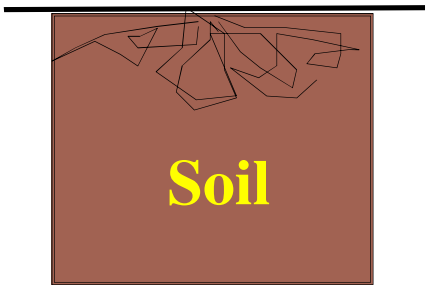
Crop & soil temperature

Irrigation

Water demand

Soil evaporation

Plant transpiration



N fertilizer

N uptake

N_2 **Fixation**

SOM N Mineralization

Residue decomposition

Organic Manure

Water & nitrate transfer

- All the processes simulated at a day-time scale
- In interaction with technical acts at the rotation scale

Issue, means and costs of genericity (i.e. flexibility)

Issue = to represent a great range of variability of productions over space and time.

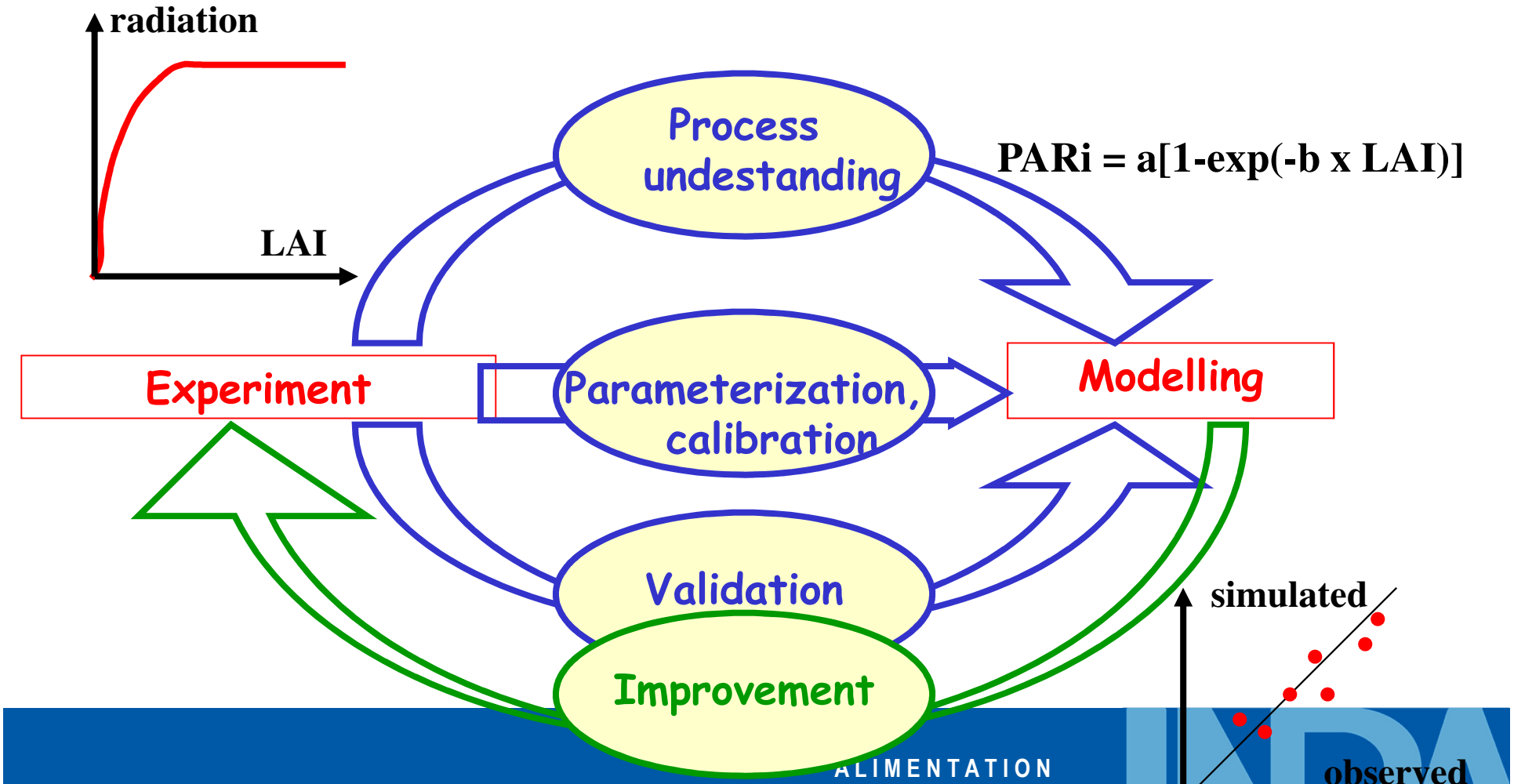
Means =

- Distinction between general and local parameters.
- Option to deal with crop or soil or practices variability.
- Functionnal (analogic) description of processes : ie RUE for DM accumulation, capacitive for water storage; mixing cells for solute transfer; solar energy and nutrients sharing by intercrops…
- STICS modularization for coupling or chaining within platforms.

Costs =

- Some parameters without biophysic meaning, need to be calibrated.
- Validity domain (robustness) depends on calibration and validation database quality.

Complementarity experiment / modelling
i.e. *In vitro & in situ vs in sillico experiment*



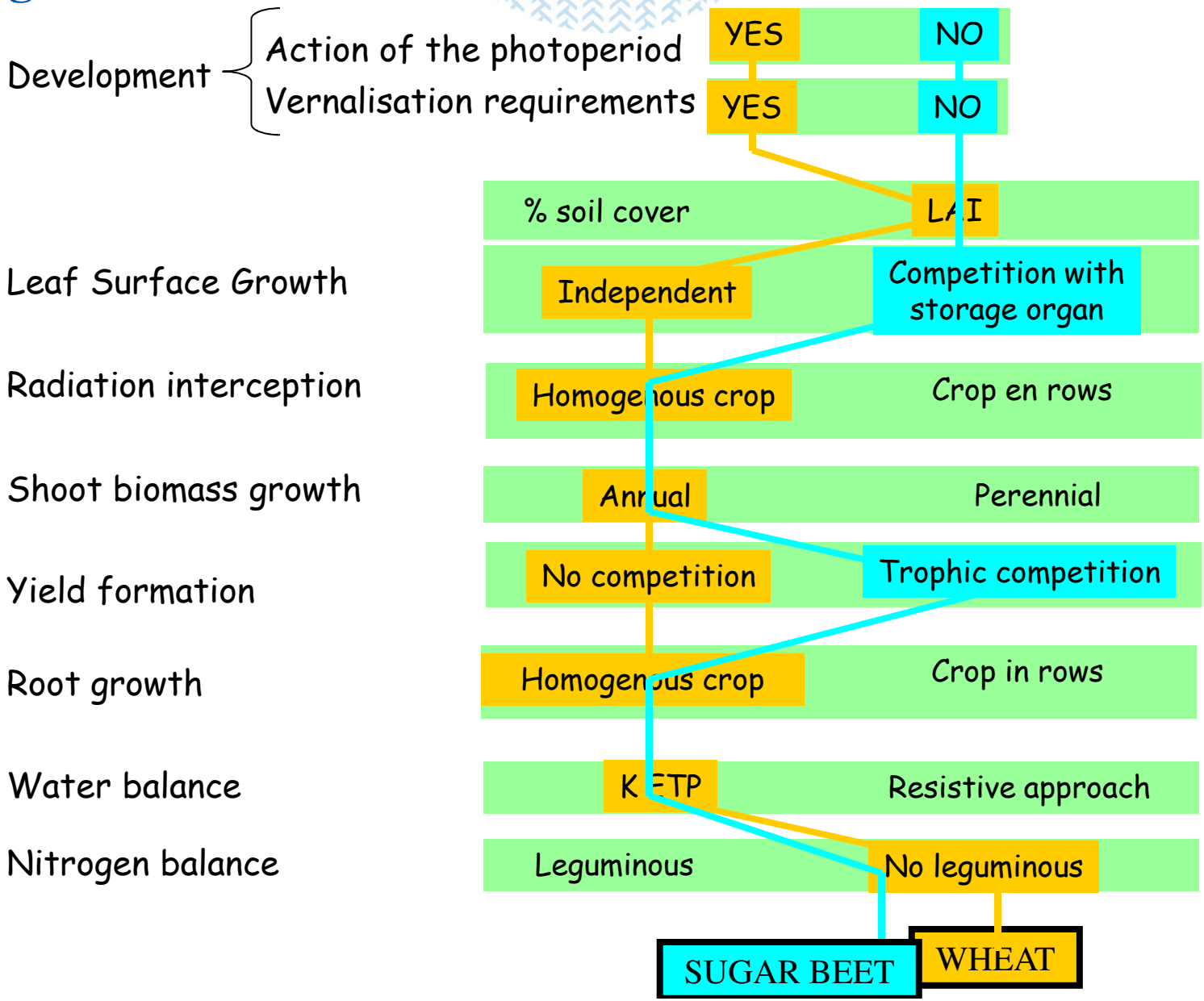
Crop generality through parameterization

Structure

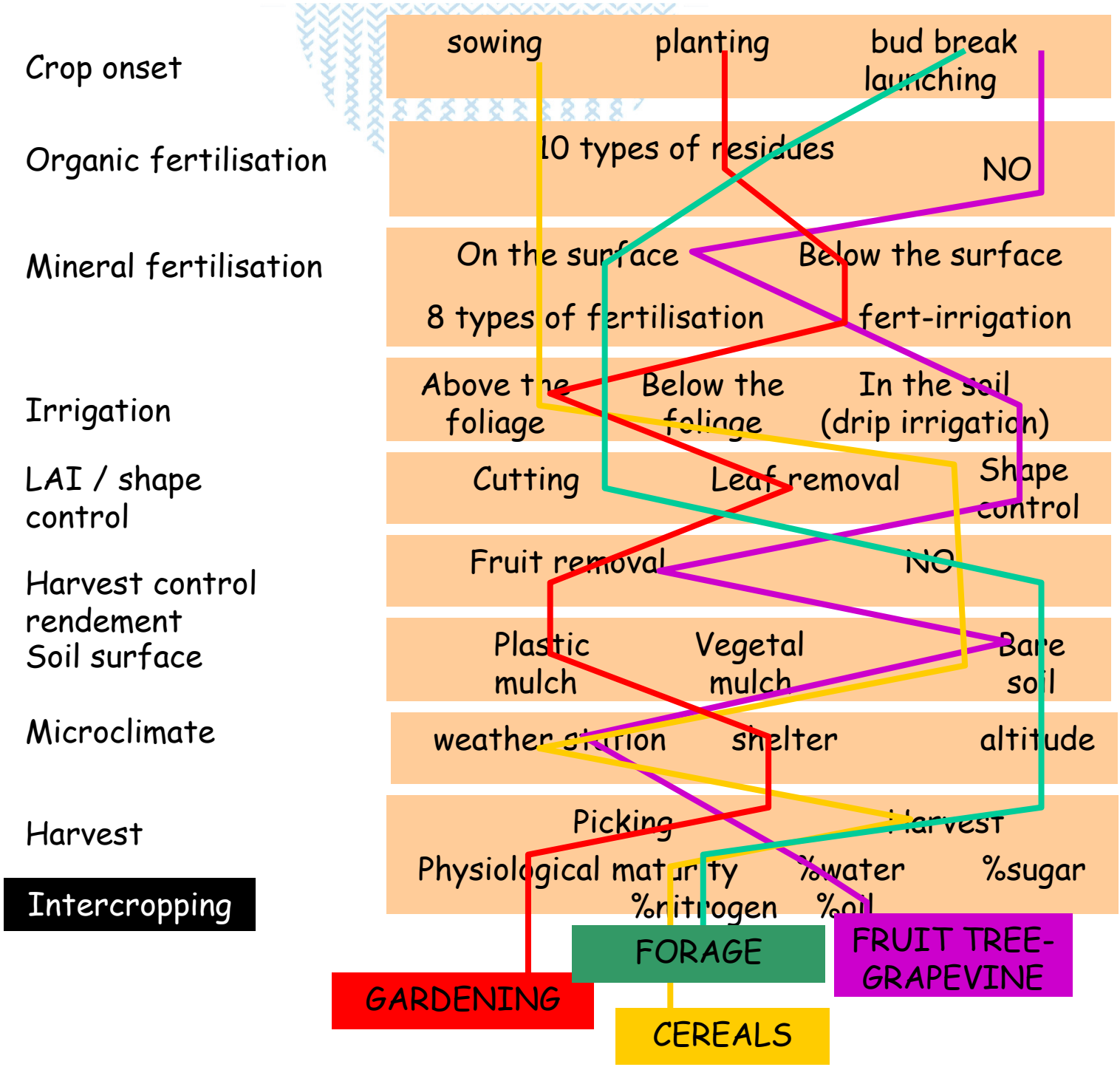
Parameter	*Wheat Shango*	*Corn Dunia*	*Banana*	*Sugar cane*	*Grapevine Grenac.*	*Syrah*
DLAIMAXBRUT (m^2 degree.day^{-1} plant^{-1})	5.10^{-4}	16.10^{-4}	200.10^{-4}	8.10^{-4}	150.10^{-4}	
CROIRAC (cm degree.day^{-1})	0.14	0.15	0.05	0.06	-	
EXTIN	0.5	0.7	0.7	0.6	-	
TDMIN(°C)	0.0	6.0	14.0	12.0	10.0	
TOPT(°C)	12-17	24-32	21.0	13-28	25.0	
TDMAX (°C)	28.0	32.0	35.0	30.0	37.0	
EBMAX (g MJ-1)	4.3	3.8	1.8	3.5	2.3	
PHOBASE (hour)	6.3	-	-	23.0	-	
PHOSAT(hour)	20.0	-	-	13.5	-	
STDORDEBOUR (degree.day)	-	-	-	-	9174	7819
DURVIEF (Q10)	205	200	1000	305	400	

Crop generality through formalization

Structure



Crop management



Types of STICS use in cropping system study or management

At the **"plot-daily" scale**, thanks to knowledge synthesis :

=> diagnosis = understand how the system works;

=> prognosis = predict the impact of changes

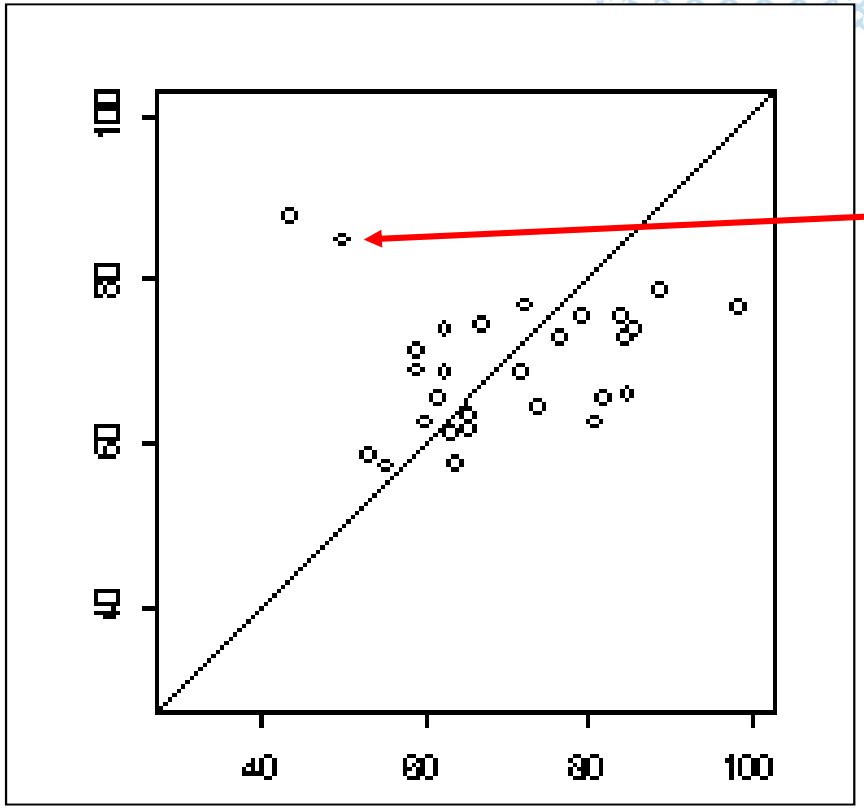
=> support decision making in complex systems

At the "**plot-crop cycle scale**" or more integrated one.

=> Test of cognitive hypothesis, possibly through model integration (either temporal or spatial or systemic)

=> Interdisciplinary dialogue between actors

diagnosis



Launay et al..

Yield accumulation predicted by model

factor "nematodes" not taken into account »

complementarity between model and measurements can make the diagnosis "default" when the model does not describe the factor responsible for the variability

Why are wheat yields stagnating in Europe? A comprehensive data analysis for France

Brisson et al. 2010.

Estimation climatic penalty of wheat yield by modeling.

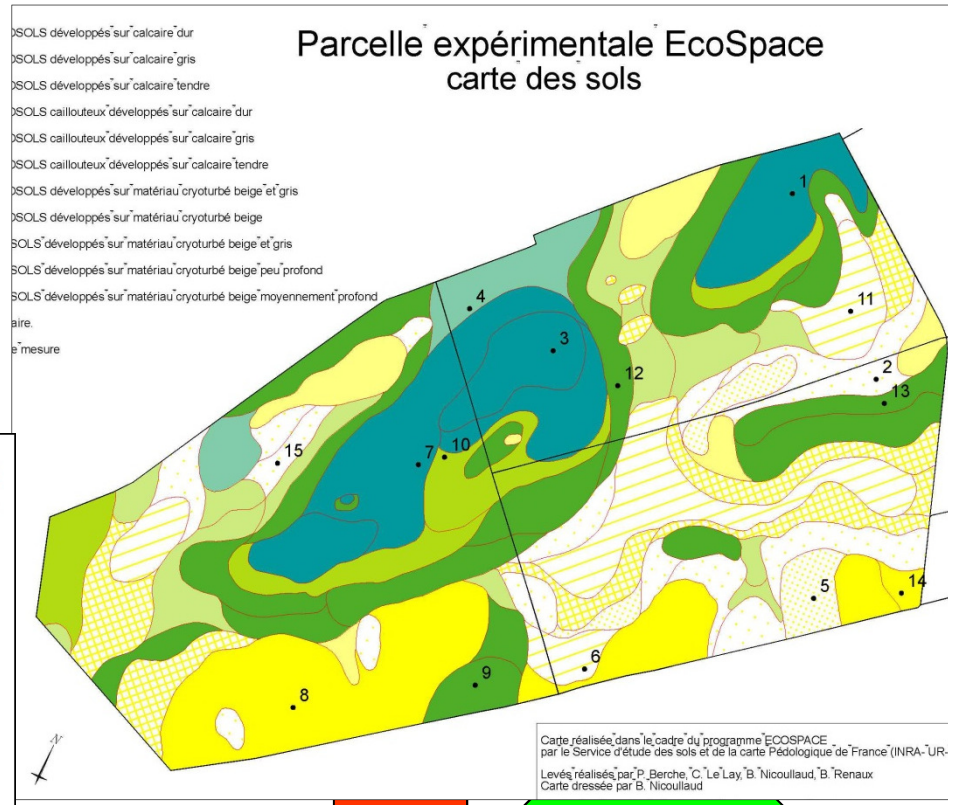
locations	Estimation of the climatic penalty of wheat yield ($t\ ha^{-1}\ yr^{-1}$) between the two periods statistically defined (year of change of slope)	
	PANORAMIX	STICS
Amiens	−0.008 (1983)	−0.074 (1998)
Bourges	−0.039 (1993)	0.000 (1997)
Dijon	−0.107 (1993)	−0.038 (1999)
Rennes	−0.002 (1993)	−0.001 (1992)
Toulouse	−0.022 (1980)	0.000 (1982)
Nîmes	−0.078 (1988)	−0.016 (1984)
Mean (std)	−0.043 (0.042)	−0.021 (0.030)

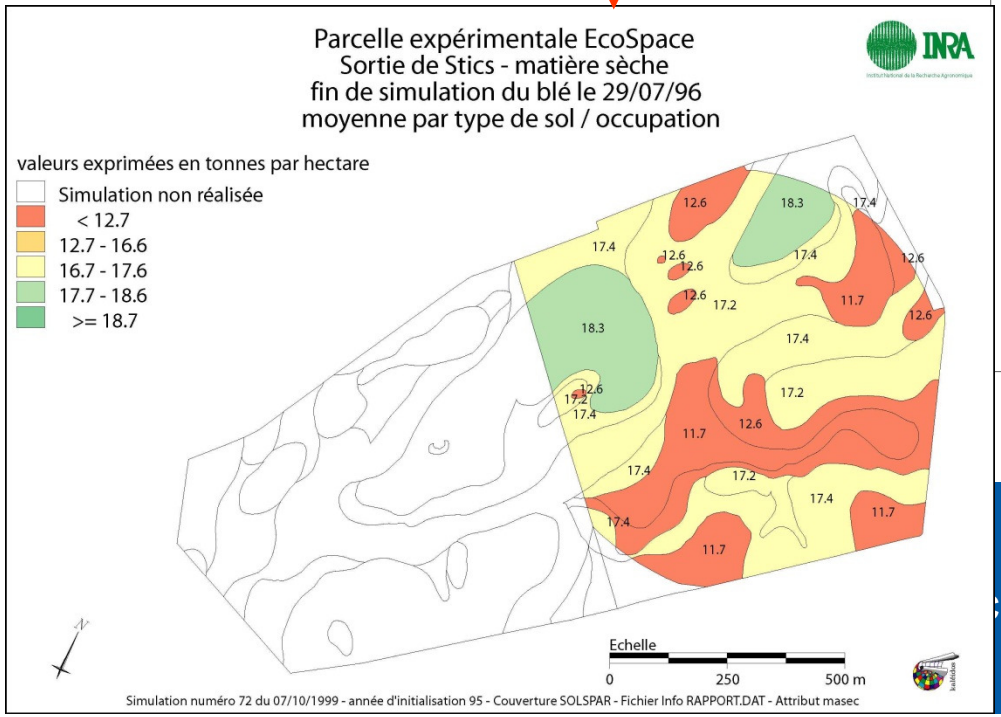
Application issues finalized and confrontation of model predictions

prognosis

intra-plot spatial variability of performance (*Nicoullaud et al., 2005*)

From the spatial variability of the system,
characterized at a given time, the final variability system can be provided by the model.

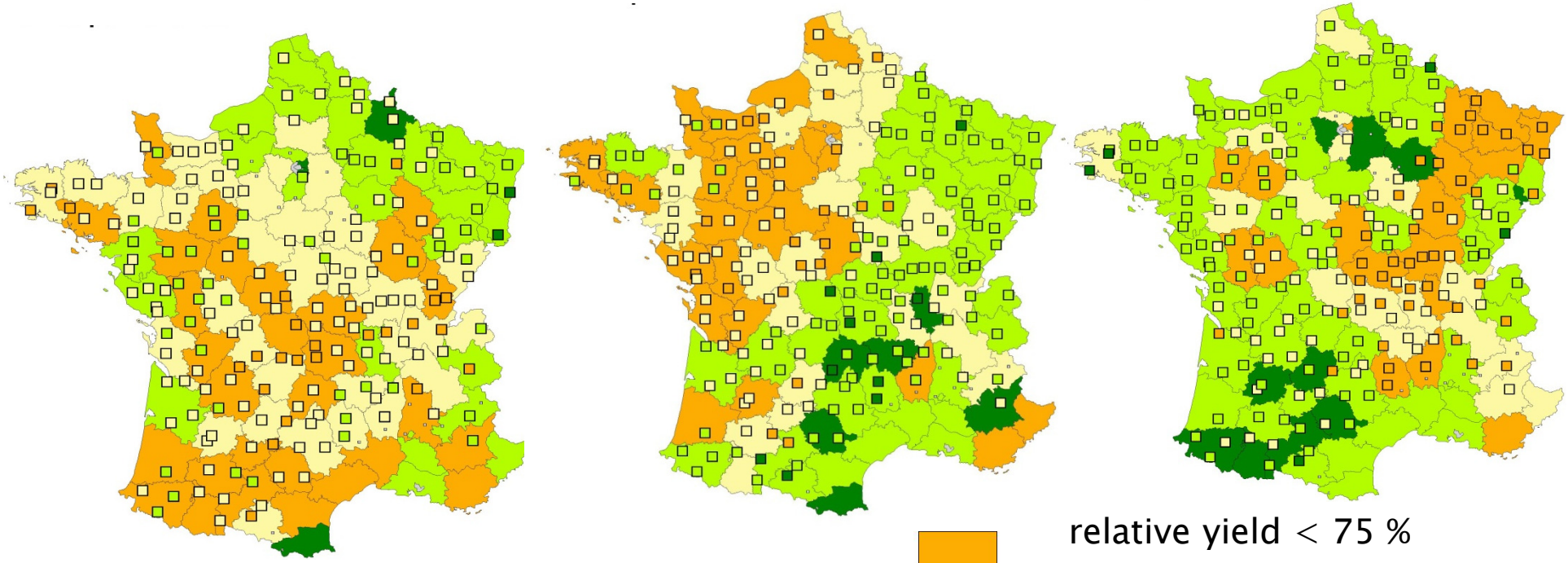




(STICS)

Evaluation of the interannual variability of production grasslands in France *(Ruget 2003)*

1989 1990 1991



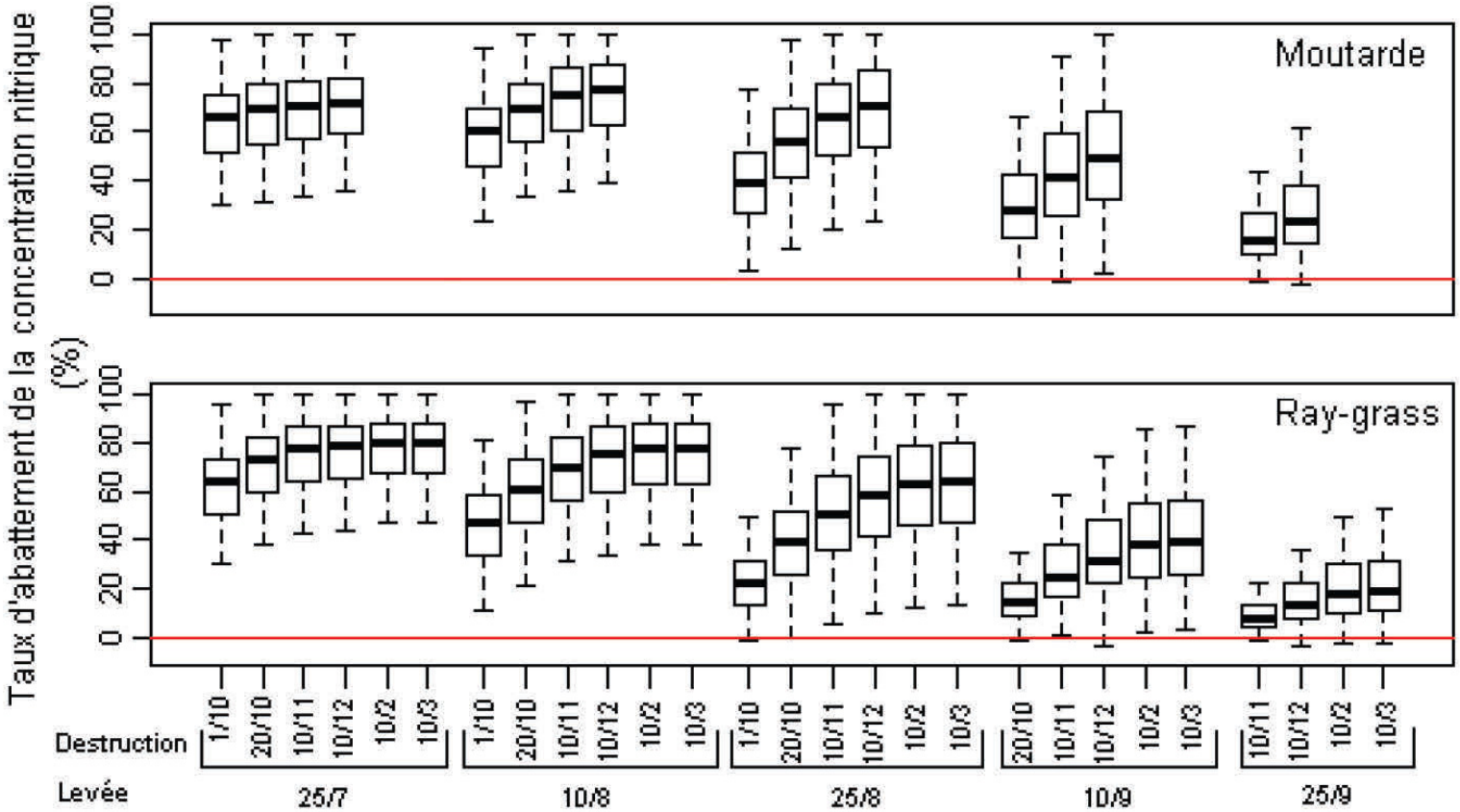
Square = forage land (ISOP)

Link with climate databases and practices

- relative yield < 75 %
- 75<relative yield < 90 %
- 90<relative yield < 110 %
- relative yield >110 %
- without data

Reduction rate of the nitrate of seeping water according to a catch crop scenario

Justes et al. 2012.

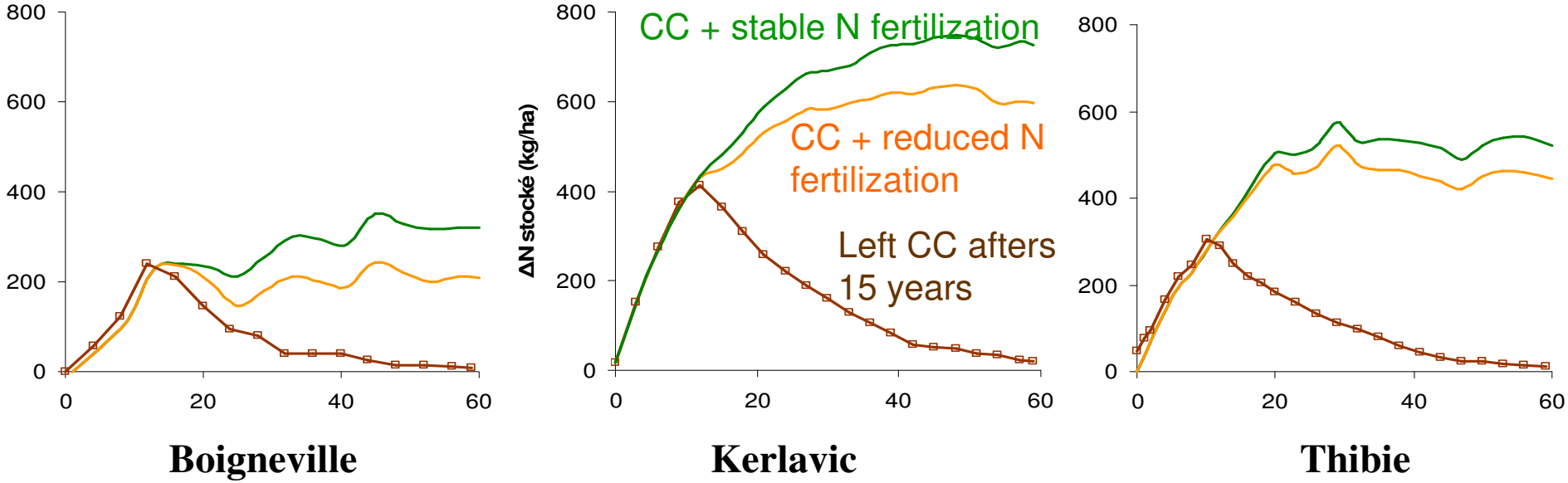


Interest of the model to predict a variable not visible.

Interest also to simulate several scenarios than reality.

cognitive tool for testing hypotheses

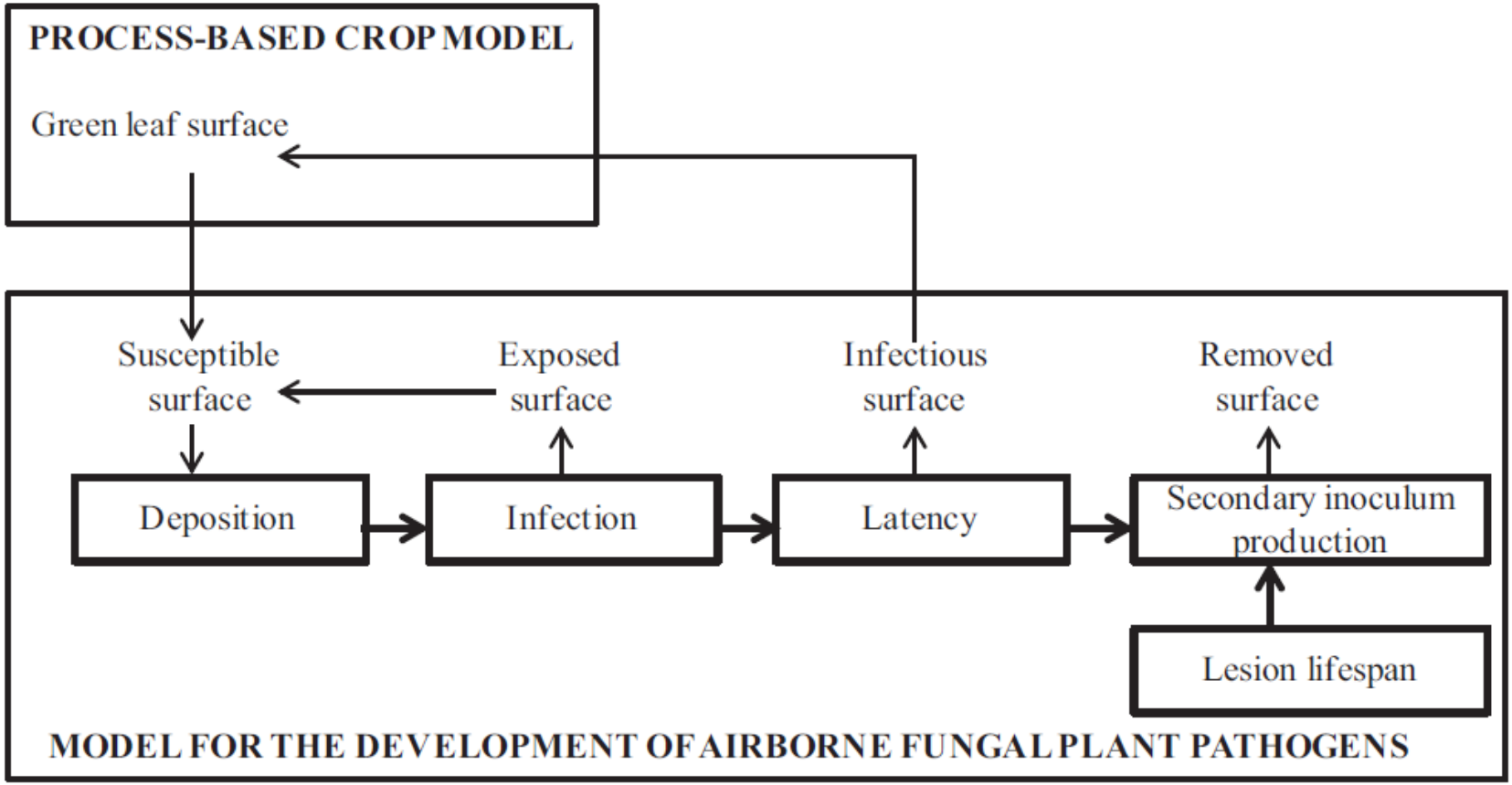
Long-term Soil N storage in various scenarios of catch crop (CC) as compared bare soil: *Constantin et al. 2011.*



Organic nitrogen storage due to CC have been experimentally observed following the introduction of CI for 15 years (Constantin, 2010). It can be simulated by STICS while it trends to stabilize at long-term.

models for the development of airborne fungal crop pathogens

Caubel et al. 2012.

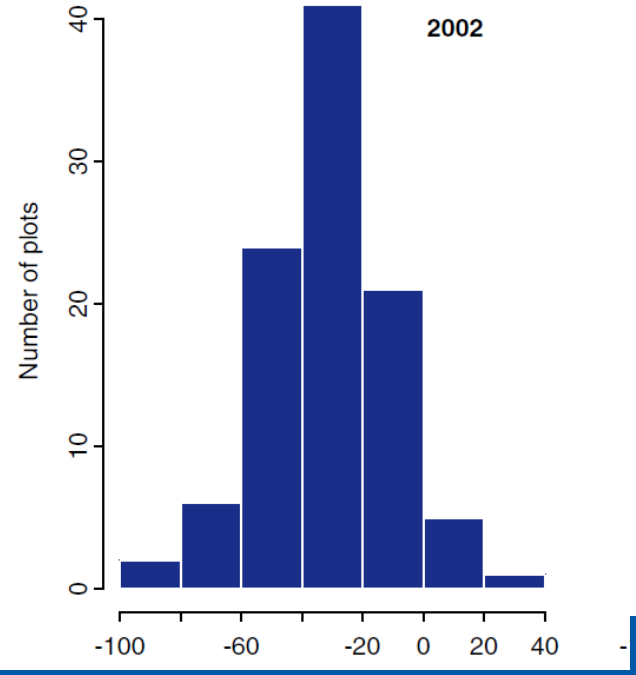


Functional approach applied to biotic processes

Supporting stakeholder decision

Epistics: A dynamic model to generate nitrogen fertilisation and irrigation schedules in apple orchards, with special attention to qualitative evaluation of the model

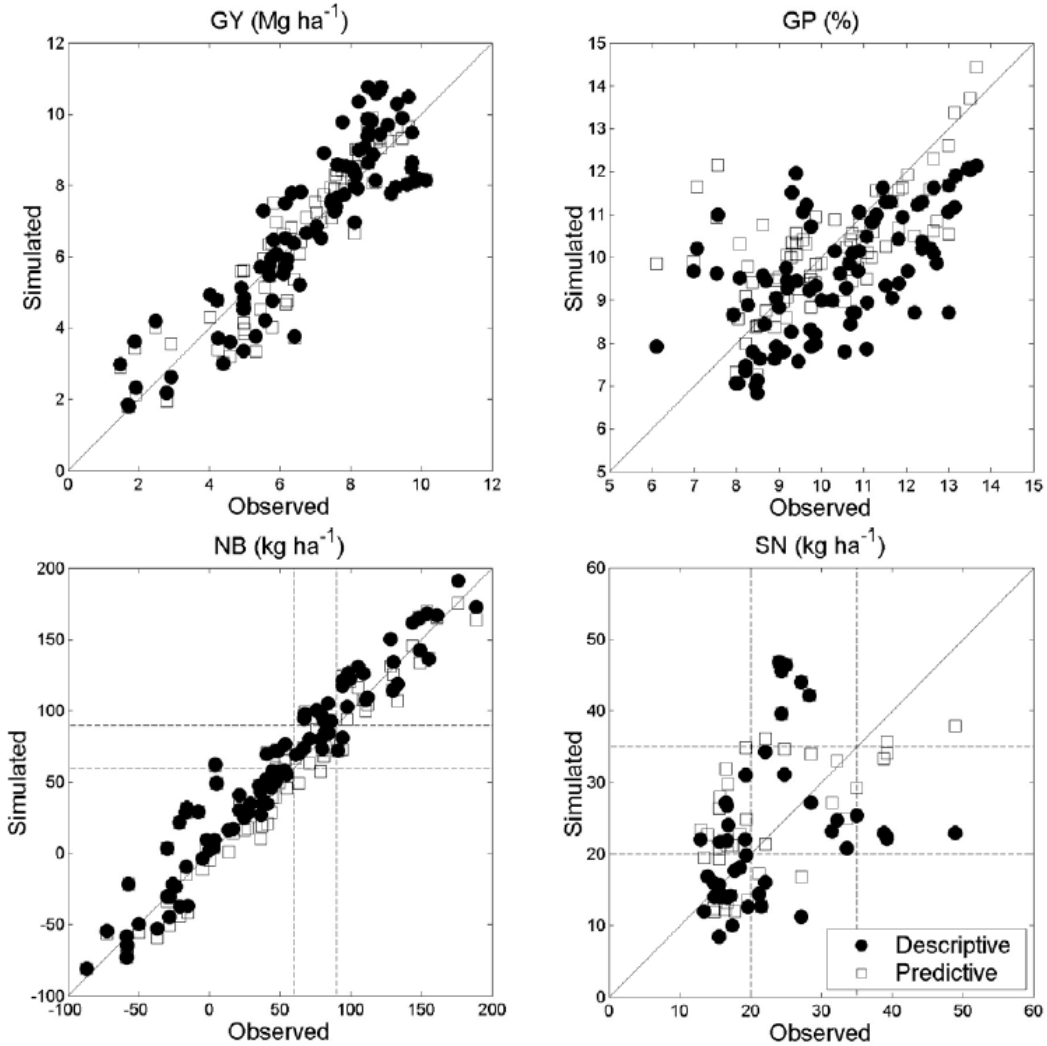
Nesme et al. 2006.



Difference (Estimated – observed) days of year of beginning of irrigation for years 2002.

Description of the decision
Confrontation with the decision
Providing the meaning of the discrepancy

Evaluation of the ability of the crop model STICS to recommend nitrogen fertilisation rates according to agro-environmental criteria



Houles et al. 2004.

Comparison of simulated and observed values of grain yield (*GY*), grain protein content (*GP*), nitrogen balance (*NB*) and soil mineral nitrogen at harvest (*SN*).

The model efficiency depends on the variable

⇒ Share views

Role in the dialogue between actors

Dialogue for STICS design or test

+ Between designers => conflict between opening to innovation and caring for robustness => distinguish 1 Standard and n versions Research (GA 2007)

+ With specialists about process & meaning of the parameters; possible incomprehension due to the STICS functional pattern.

+ With users, to check STICS against new context or rationale, => anticipate experiments, pool & manage databases.

Dialogue about the cropping system assessment

+ Between agronomists => about driving factors of the crop. system.

+ With environmental or economic scientists ; => variables in common, driving factors, scale effects in global system (agricultural, hydrological, ecosystem ..) where the cropping system interacts.

Thank you for your attention

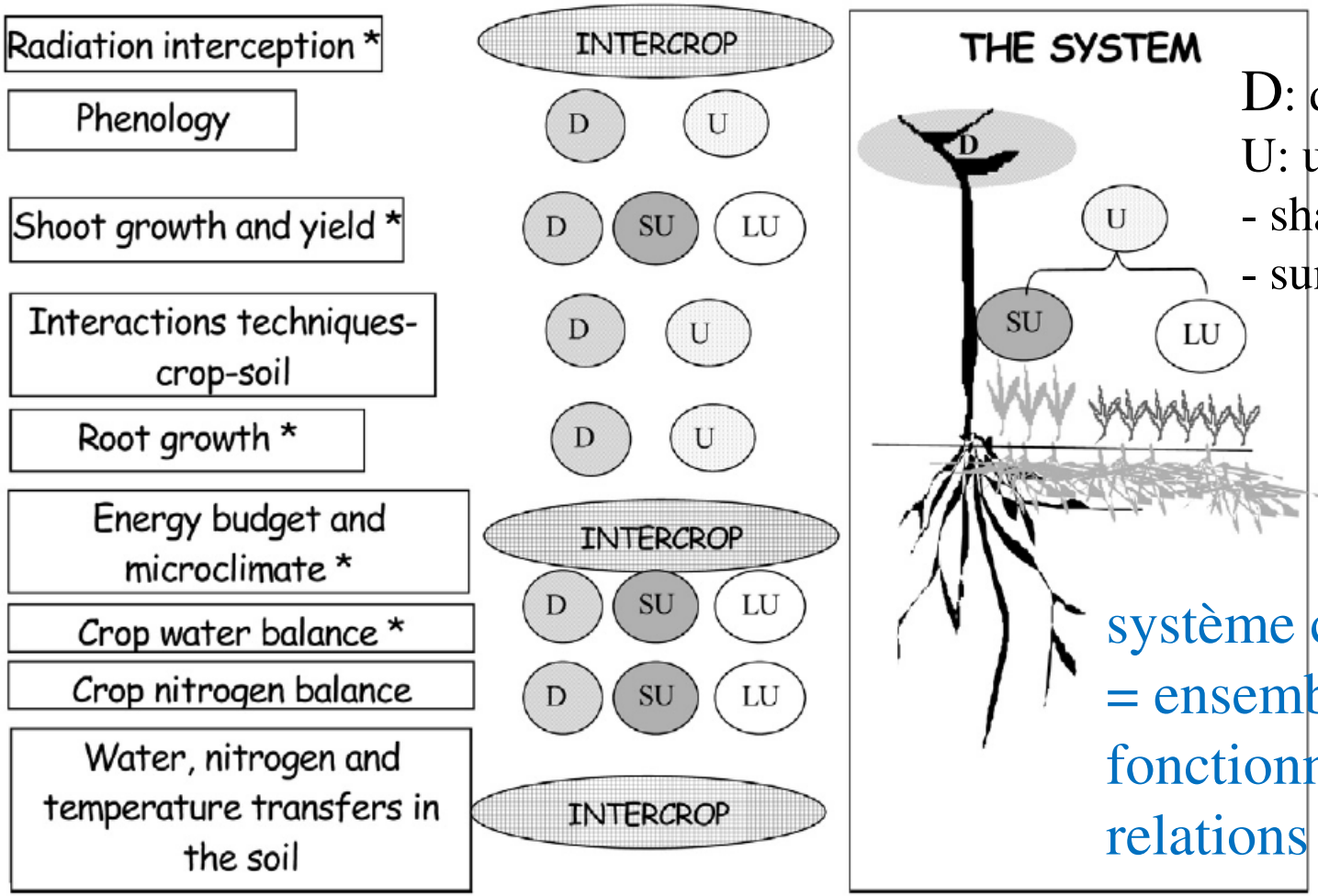
Thank you Nadine

Merci pour votre attention et votre compréhension / absence dia en français

outil cognitif de test d'hypothèses

Adaptation of the crop model STICS to intercropping. Theoretical basis and parameterisation

Brisson et al. 2004.



D: dominant crop;
U: understorey crop
- shaded part: SU
- sunlit part: LU

système complexe
= ensemble d'unités fonctionnelles en relations

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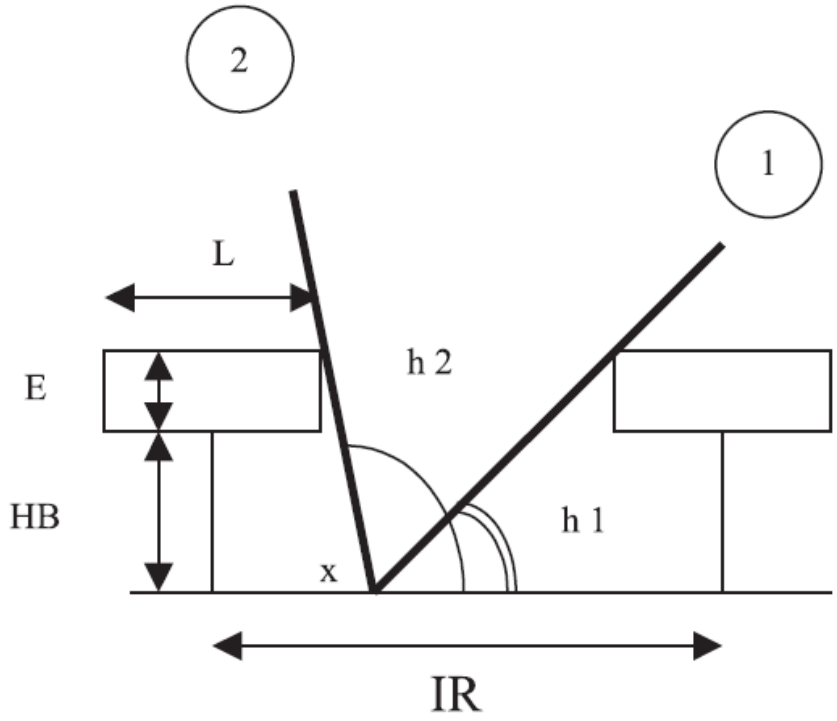


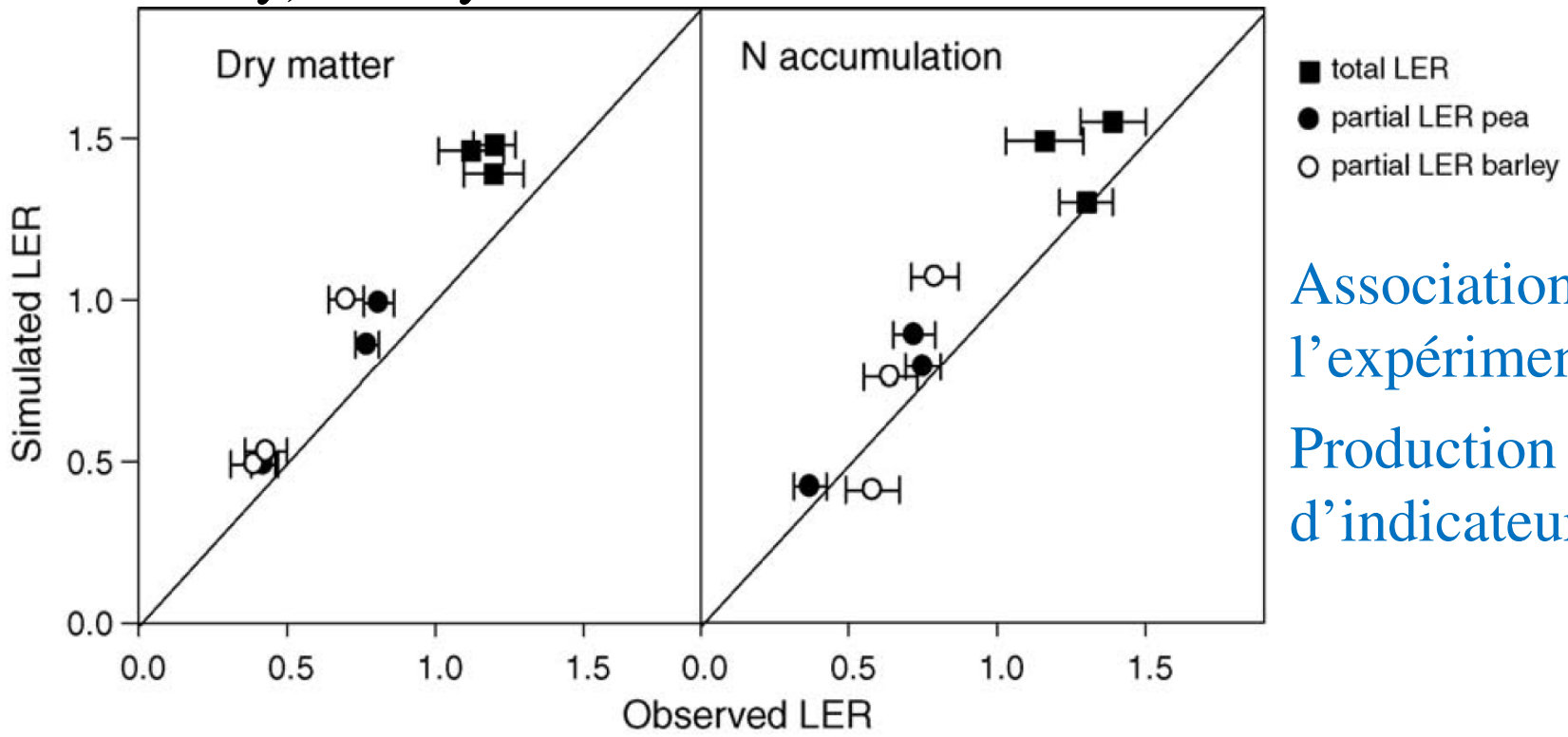
"Simplified representation of the plant canopy and the principles used for calculating daily radiation received by the interrow (IR) »

description physiquement basée et simplification

crop growth and N accumulation in pea–barley intercrops

Corre-Hellou et al. 2009.

Simulated and observed total and partial LER values (LER pea and LER barley) for dry matter and N accumulation



Association à l'expérimentation

Production d'indicateurs